Pool-ball localisation and recognition

Computer Vision (CSU44053) | Manon Yiqin Galian Ng (23375206)

1. Aim and overall approach

The system locates pool balls in a stationary-camera photograph, maps their centres into table coordinates, and assigns each an identity: cue or number 1–15. The principal difficulties are incomplete outlines in touching groups and appearance changes caused by orientation, highlights and shadow. A white-facing stripe can resemble the cue; a solid ball's number patch can resemble a stripe.

Localisation combines circular Hough detection with the supplied table geometry. Recognition combines two-cluster colour segmentation, reference colour histograms and a joint assignment of identities. Only `AllBalls.jpeg` supplies labelled colour-training examples. All six table photographs use the same final settings. The processing uses the smoothing, Hough, clustering, histogram and evaluation methods introduced in the course [1].

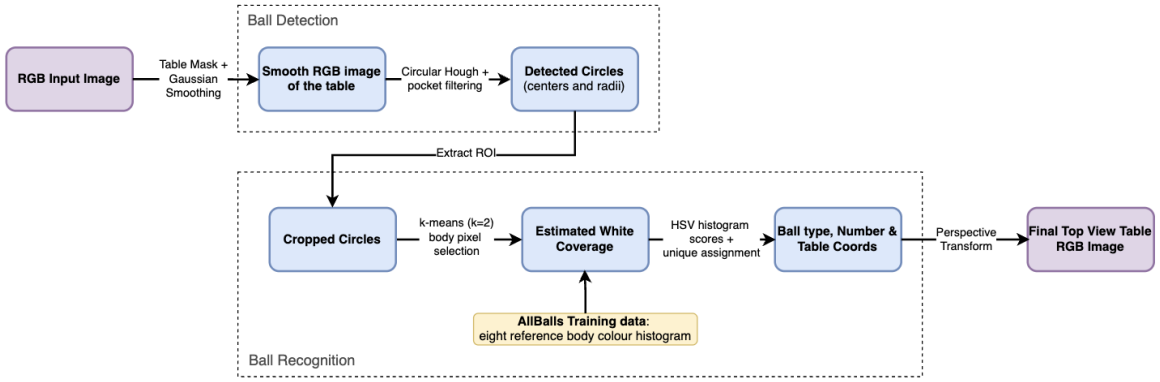


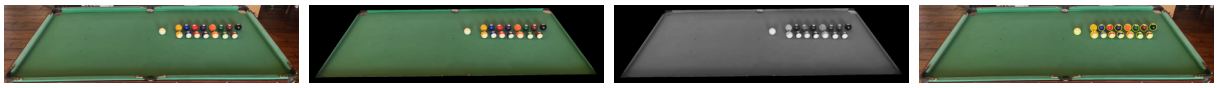
Figure 1: Overview of the pool-ball localisation and recognition pipeline.

2. Localising the balls

2.1 Restricting the image and finding circles

The camera position is fixed across the supplied photographs. I therefore used the four supplied table corners to create a quadrilateral mask: top-left $(337, 35)$, top-right $(1570, 21)$, bottom-right $(1888, 451)$ and bottom-left $(38, 472)$, written as (x, y) in the original image. This prevents edges in the surrounding room from becoming candidate balls. The masked image is converted to greyscale and blurred with a 5×5 Gaussian filter ($\sigma = 1.5$ pixels), which reduces small variations that would otherwise produce distracting edge responses.

A gradient-based circular Hough transform searches the smoothed image for circular outlines. Its output is a list of centres and radii, not a list of confirmed balls. This distinction matters: a pocket edge or the white region inside a striped ball can also generate a circle. I therefore treat the Hough output as a starting point and apply filtering and recognition afterwards.



Original image | Table mask | Greyscale + blur | Retained circles

Figure 2: Localisation stages on `AllBalls`. The colour image is retained separately for recognition.

2.2 Choosing detection parameters

Rather than setting the Hough thresholds only by eye, I searched 480 combinations on `AllBalls.jpeg`. The search varied accumulator resolution, minimum spacing between centres, edge threshold and circle-voting threshold, with the radius range held at 15–30 pixels. I ranked configurations by matched detections, then false positives and centre error. The selected settings were an accumulator resolution 1.5 times coarser than the image, minimum centre spacing 18 pixels, upper edge threshold 75 and circle-vote threshold 35. Those values were fixed for the six table images.

Lower vote thresholds admit weaker boundaries but also more false circles. Minimum centre spacing suppresses duplicate responses, but can also suppress a second ball in a crowded group. These effects explain the need to measure both missed balls and false positives.

2.3 Using the table geometry

Six pockets have known approximate positions on the table. I transform each Hough centre to table coordinates, then reject candidates within 25 table units of a pocket centre. This removes a predictable source of false circles, although a real ball close to a pocket could be lost by the same rule.

A planar homography maps the supplied corners to $(0, 0)$, $(1433, 0)$, $(1433, 682)$ and $(0, 682)$ in a 1434×683 rectangle. Thus one coordinate unit is approximately one millimetre, with x increasing rightwards and y downwards. The transformation has the form $\lambda(x_t, y_t, 1)^T = H(x, y, 1)^T$, where four point correspondences determine H up to scale. Only the *detected centre* is transformed for the final coordinates. Detection and colour recognition remain in the original photograph: warping the image beforehand could deform circular ball outlines, making Hough detection and circular crops less reliable. A rectified full-table image is produced separately for displaying results.

3. Recognising colour and ball type

3.1 Why a crop needs more than a colour threshold

For each retained circle I take a square crop from the original colour photograph and apply a circular mask. This keeps the image's actual colour values while excluding most surrounding cloth. A ball can contain its body colour, a white number patch, bright reflections and shadow, so simply averaging all pixels is unreliable. The recogniser instead asks two questions: how much of the crop looks white, and which coloured ball does the remaining part resemble?

To estimate white coverage, I cluster the circular crop's BGR pixels into two groups using K-means. Each crop uses three initialisations with the same random seed; iteration stops after 50 steps or a centre change below 0.2 colour units. Each cluster is represented by its

mean colour and its proportion of the circular crop; these proportions, rather than direct pixel thresholds, determine white coverage. The cluster colour is converted to HSV: it is considered *light* when $V \geq 170$ and $S \leq 165$, *dark* when $V \leq 90$, and otherwise *coloured*. Dark counts as non-light, because purple, maroon and black bodies should not be mistaken for white. The initial rule assigns cue when at least 85% is light, solid when at least 70% is non-light, and stripe to the remaining mixed cases.

This is a useful first estimate, not a final decision. The visible proportion of white changes with ball orientation, and K-means can merge a pale body colour with a white highlight. In particular, some striped balls are initially mistaken for the cue.

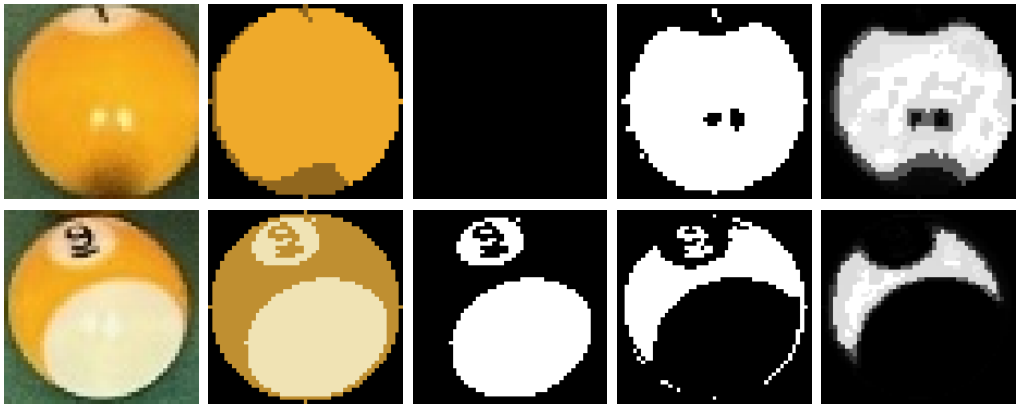


Figure 3: Solid 1 (top) and striped 9 (bottom). The same body colour appears in both balls, but their white coverage differs; the final column visualises a yellow histogram response.

3.2 Matching the coloured part of a ball

The original recogniser assigned colour using fixed HSV ranges. That worked when a large, well-lit body-colour patch was visible, but broke down when the cluster representative fell near a threshold or a highlight changed brightness. I therefore compared the whole distribution of colour within the body region, rather than relying on one representative hue. I used eight reference histograms learned from `AllBalls`: yellow, blue, red, purple, orange, green, maroon and black. Each model combines the solid ball with its striped partner, so colour recognition does not depend on getting the type correct first. The cue ball is excluded from these models.

For a new crop, I erode the circular mask slightly (a 3×3 elliptical kernel) and remove light pixels before measuring body colour. This avoids allowing a large white stripe or the edge of the green table to dominate the comparison. If fewer than 12 usable body pixels remain, colour evidence is considered insufficient rather than inventing a confident label. Each reference and query uses an $18 \times 8 \times 8$ HSV histogram. Hue spans $[0, 180)$ and saturation/value span $[0, 256)$. Pixels with BGR chroma below 15 have their hue set to zero, reducing unstable hue variation near grey. A $[1, 2, 1]/4$ filter smooths each histogram axis, wrapping hue and replicating the saturation/value endpoints. Histograms are normalised to sum to one; solid and stripe training examples contribute equally to their shared colour model. Similarity is $s = 1 - \sqrt{1 - \sum_b \sqrt{p_b q_b}}$, where p_b and q_b are the normalised reference and query bin masses. Higher similarity means closer colour distributions [1]. The displayed backprojections illustrate where colours match; they are not the decision score.

4. Making one consistent prediction per ball

Independent crop classifications can assign the same identity to several detections. I therefore score all 16 identities before choosing a consistent set. Let w be a crop's light fraction. Expected intervals are $[0, .30]$ for solid, $[.30, .85]$ for stripe and $[.85, 1]$ for cue. For interval I_t, type compatibility is

$$g_t(w) = \exp\left[-\frac{d(w, I_t)^2}{2(0.10)^2}\right],$$

where d is zero inside the interval and the distance to its nearest endpoint outside. A numbered identity receives its colour-histogram similarity multiplied by the appropriate solid/stripe compatibility. The cue receives $wg_{\text{cue}}(w)$. These are heuristic ranking scores, not probabilities.

The Hungarian assignment method [2] maximises the sum of scores, with each identity available at most once. Separate dummy identities allow any number of `Unknown` predictions; missing real identities need not be assigned. Numbered scores require at least 12 body pixels. An identity is eligible only with usable colour evidence or an initially white classification, and score above 10^{-9}; dummy scores are 10^{-9}. This is a best-effort policy: it can accept weak alternatives and does not provide a reliable confidence-based rejection of false detections.

Two small corrections address cue/stripe confusion. If a crop was initially called white, I enlarge its analysis radius by one pixel and repeat the colour measurements; the extra rim sometimes reveals a stripe's coloured edge. If at least two detections were initially white candidates, the one with the largest *original Hough radius* is reserved as the cue, while the others must take a number or `Unknown`. In Table1 the full cue had radius 24.15 pixels, compared with 17.55 pixels for a detected white patch. I also tried correcting radius for image depth, but it produced exactly the same predictions on the six images. I kept the simpler raw-radius rule and discuss its limitations below.



Full cue (24.15 pixels) White patch (17.55 pixels)

Figure 4: Why Hough radius can help resolve two initially white candidates (Table1). This is a heuristic, not a guarantee of correct cue recognition.

5. Evaluation setup and measurement

5.1 Images, annotations and matching

The supplied material consists of `AllBalls` and six table photographs. `AllBalls` supplies colour-training examples and Hough parameter selection. Tables1–6 contain 90 annotated positions in total. Evaluation uses the project's working annotation copy, which already includes coordinate/identity corrections and re-estimated centres relative to the original

supplied file. Both copies are retained; the reported scores are conditional on these revisions, whose justification must be distinguished from detector performance. Predictions are not manually corrected before measurement.

For localisation, predicted centres are matched one-to-one to annotated centres after both are expressed in table coordinates. A pair is allowed when it lies within 15 table units. Matching prioritises the number of valid pairs and uses proximity to resolve alternatives; it does not necessarily minimise the global sum of distances. From the matches I calculate TP (matched detections), FP (unmatched detections) and FN (unmatched annotations):

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}, \quad F_1 = \frac{2PR}{P + R}.$$

I also report mean distance between matched centres, and identity accuracy over the matched balls. An `Unknown` on a real ball is counted as an identity mistake. Finally, the number of correct identities divided by *all* annotations gives a more demanding end-to-end measure, which also penalises missed balls.

Result images show predicted labels at detected centres. Visual inspection complements the metrics: a matched centre can still have the wrong identity, while an aggregate score can conceal errors in a crowded group.

Recognition rules were revised after inspecting Tables1–6. Although table annotations enter only the evaluation stage of each run, this feedback influenced method selection. The results are therefore *development-set measurements*; training colour histograms only on `AllBalls` does not make them an independent test of generalisation.

5.2 Detection and recognition results

Table 1: Results of the selected experimental configuration. Position error is in table units (approximately mm). Identity accuracy is measured only on matched detections.

Image	GT	TP	FP	FN	Mean error	Correct identity
Table1	16	14	1	2	7.68	11/14
Table2	16	16	1	0	3.58	16/16
Table3	16	15	0	1	3.12	15/15
Table4	15	14	0	1	3.89	14/14
Table5	14	14	0	0	5.83	13/14
Table6	13	13	0	0	5.76	11/13
All	90	86	2	4	4.91	80/86

The detector found 86 of 90 annotated balls with two false positives: precision 97.73%, recall 95.56% and F_1 96.63%. The pooled mean centre error was 4.91 units, approximately 11.4% of a ball diameter. Recognition was correct on 80/86 matched detections (93.02%); end-to-end identification was 80/90 (88.89%). All six genuine cues were recognised. These totals pool counts across images; the mean error weights each image by its number of matches.

Table2 detects and identifies all 16 annotated balls, but also produces a false circle. Tables3 and 4 correctly identify all matched balls, while each misses ball 10. Table5 has full localisation recall and one identity error; Table6 has full recall and two identity errors. Table1 combines two misses, a false detection, three identity errors and the largest mean

position error. Crowding affects localisation first; unfavourable colour/white coverage can defeat recognition even when localisation succeeds.

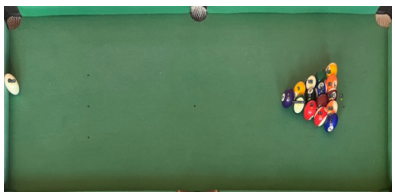


Table1: crowded group

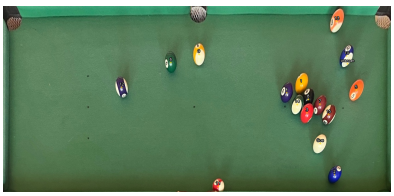


Table2: 16 correct identities

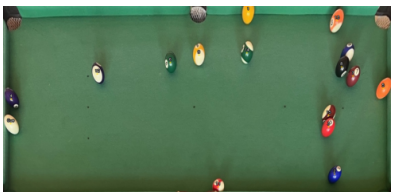


Table3: ball 10 missed

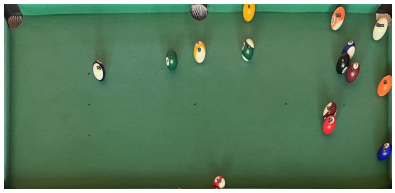


Table4: ball 10 missed

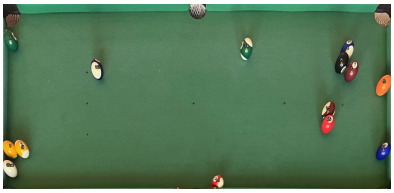


Table5: one identity error



Table6: two identity errors

Figure 5: Rectified results for all six photographs. Labels are predictions at transformed Hough centres; a correct identity count does not exclude extra detections.

6. What changed during experimentation?

Table 2 keeps all 86 matched detections fixed and changes recognition only. The main change is to treat the initial type classification as evidence that later colour and assignment stages can revise.

Table 2: Incremental recognition comparison on the same 86 matched detections.

Recognition configuration	Correct identities
Two-cluster K-means with fixed colour rules	68/86
Add HSV reference histograms	72/86
Add soft type scores and one-to-one assignment	75/86
Enlarge initially white ROIs by one pixel	79/86
Reserve the largest original-radius cue candidate	80/86

The histogram model improved recognition without changing how many balls were detected: it replaced a brittle colour boundary with a comparison to examples of the same materials and lighting. Assignment then helped in cases where two candidates claimed one identity. Padding was especially helpful when Hough had captured a stripe's inner white area. The one-pixel change seems small, but for a tightly cropped circle it can expose enough colour to change the ranking of several identities. Cue reservation fixed one further mistake in Table1, but was not the source of the larger improvement.

I compared the raw-radius rule with the earlier depth-adjusted alternative. Both selected the same cue candidates in Tables1, 3 and 6 and produced identical final predictions. Since the fitted depth relationship came from a narrow band of training positions, keeping it would have added an extrapolation assumption without any measured gain. The simpler rule is easier to reproduce and explain; it does not, however, solve perspective in general.

Table 3: Cue-size ablation with all other stages held fixed.

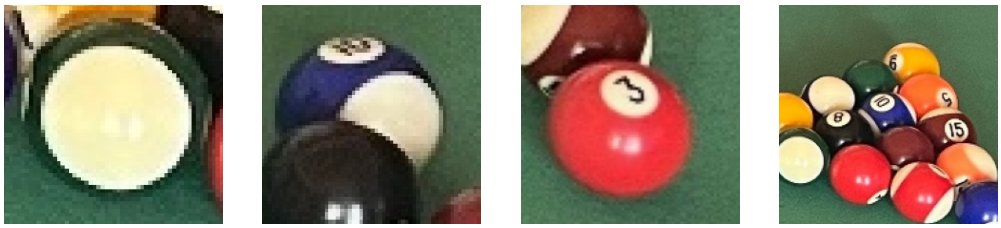
Rule	Correct identities	Correct cues
No reservation	79/86	5/6
Largest raw radius	80/86	6/6
Largest depth-adjusted radius	80/86	6/6

7. Failure cases and rejected alternatives

7.1 Where the selected approach still fails

Most localisation errors appear where balls overlap. Table1 misses balls 6 and 7 in a crowded group; ball 10 is missed in both Tables3 and 4. When the outline is interrupted by another ball, the Hough evidence weakens, and nearby candidates can compete under the minimum-centre-spacing rule. This is a detection failure: no subsequent colour classifier can recover an object that was not detected.

Recognition errors tell a different story. In Table1, the real cue crop includes some green cloth, while stripe 14 provides only eight usable coloured pixels after padding, below the required twelve; stripe 14 remains `Unknown`. Table1 also confuses solid 5 and stripe 9. Tables5 and 6 mistake some blue stripes for solid 4, and Table6 mistakes solid 3 for stripe 11. These failures are plausible from the crops: the amount of visible white varies with orientation, and an imperfect circle includes neighbours or background. Uniqueness is helpful overall but can also propagate a mistaken assignment.



Stripe 14: Unknown Blue stripe error Solid 3/stripe 11 Occluded balls (Table1)

Figure 6: Representative mistakes, showing the original detected crops and a crowded localisation region. The circles are predictions, not corrected ground-truth boundaries.

7.2 Why more elaborate segmentation did not win

I tested several ways of recovering or separating difficult ball regions. Watershed segmentation started from Hough markers but could not recover a ball when the required marker was missing. A more permissive cluster-first/local-Hough variant recovered three annotations, yet lost three former matches and raised false positives from 2 to 16; its detection F_1 fell from 96.63% to 89.58%. Connected-component analysis examined white-region connectivity and boundary contact to distinguish stripes from number patches, but reduced correct identities from 72 to 69 with the same histogram models. A chamfer edge-based search sometimes improved one crop's boundary but scored 77/86 overall, below selective padding's 79/86. Direct pixel-level HSV counting scored 79/86 in the final combination versus 80/86 for K-means.

These comparisons distinguish a plausible boundary from a correctly identified physical ball. A cleaner segmentation can still create extra candidates or alter a crop enough

to worsen recognition. I therefore judged alternatives by detection and identity metrics rather than visual appearance alone. The final pipeline retains simpler, fixed rules, but its advantage here still needs confirmation on unseen scenes.

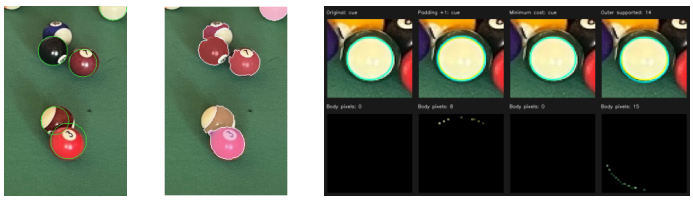


Figure 7: Examples from rejected alternatives: a crowded Hough region, watershed segmentation and edge-based chamfer crop refinement. Local improvements did not improve the total scores.

8. Limitations, conclusion and resources

The annotations give labelled centres, not exact outlines, so they cannot establish mask accuracy. The homography assumes the table plane, while visible ball centres lie above it; this can introduce systematic position error. A 15-unit tolerance is about 35% of the 43 mm diameter. A tolerance-sensitivity curve and inspection of the largest errors would distinguish precise localisation from detections that only just match. Annotation revisions also need independent justification; a missed Hough detection alone is not evidence that a supplied annotation is wrong.

The fixed camera, similar lighting and repeated spatial arrangements all favour this approach. Cue selection by the largest raw radius could fail if a distant cue is smaller than a nearby white-facing stripe, or if a spurious detection is unusually large. The experiment does not test those cases, absent-cue scenes or substantial viewpoint changes. Its histogram models are also likely to depend on the camera's colour response and lighting; adding new training images or normalising illumination would be necessary for wider deployment. Unique assignment assumes a standard, non-duplicated set of numbered balls, which is appropriate for this table but would not generalise to arbitrary coloured objects. Moreover, because Tables1–6 informed later recognition decisions, genuinely unseen images are needed before making a generalisation claim.

Histogram matching, joint assignment and cue-ambiguity corrections improve recognition on these scenes. Touching balls, inaccurate crops and orientation-dependent white coverage remain the main weaknesses. Further development should test new photographs with varied cue depths and close contacts. Consolidated processing reproduces the selected results and recognition ablations.

AI and materials acknowledgement. The module supplied the photographs, geometry and original annotations. OpenAI ChatGPT/Codex assisted with experimental design, C++ implementations, debugging, comparisons, figures and drafting. For this revision, it consolidated the reported methods, added automatic batch evaluation, verified outputs against saved experiments, and revised the report. The author must review all assistance and claims before submission.

References. [1] K. Dawson-Howe, CSU44053 course notes: *Images, Edges, Histograms, Region Segmentation 1–2, Learning & Evaluation*; based on *A Practical Introduction to Computer Vision with OpenCV*, Wiley (2014). [2] H. W. Kuhn, "The Hungarian method for the assignment problem," *Naval Research Logistics Quarterly*, 2, 83–97 (1955). [3] L. Vincent and P. Soille, "Watersheds in digital spaces," *IEEE TPAMI*, 13, 583–598 (1991). [4] G. Borgefors, "Hierarchical chamfer matching," *IEEE TPAMI*, 10, 849–865 (1988).